

## WEISSFLUHJOCH, Switzerland

WMO: 067800

Lat: 46.8333N

Lon: 9.8064E

Elev: 2667

StdP: 73.13

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |       |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|-------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |       | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB  | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)   | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -20.5      | -18.4 | -33.2                         | 0.2 | -6.6 | -30.0 | 0.3 | -6.7 | 18.3                  | -10.6 | 15.6 | -9.3 | 6.1                   | 0    | 0.909 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 6.0                    | 15.2            | 8.7  | 13.6 | 8.2  | 12.2 | 7.6  | 9.9                 | 13.4 | 9.0 | 12.4 | 8.2 | 11.3 | 3.5                  | 0    |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 8.4                             | 9.5 | 10.9 | 7.4 | 8.9 | 10.1 | 6.6 | 8.4 | 9.4  | 36.6          | 13.5 | 34.0 | 12.4 | 31.9 | 11.4 | 13.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 14.8          | 12.5 | 10.8 | DB                         | -22.5 | 19.2               | 2.5 | 2.0                                                | -24.3 | 20.7       | -25.8 | 21.9       | -27.2 | 23.0       | -29.0 | 24.5 | (4) |
| (5)               |      |      | WB                         | -22.7 | 11.7               | 2.5 | 1.1                                                | -24.5 | 12.5       | -25.9 | 13.1       | -27.3 | 13.7       | -29.1 | 14.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -1.3   | -8.5  | -8.9  | -6.7  | -3.5  | 0.0   | 4.8   | 6.9   | 6.9   | 3.4   | 0.9   | -3.9  | -7.0 |
|                                                                                  | DBStd       | 7.19   | 4.55  | 5.11  | 4.57  | 3.93  | 3.92  | 4.28  | 3.94  | 4.03  | 4.27  | 4.73  | 5.02  | 4.88 |
|                                                                                  | HDD10.0     | 4142   | 573   | 530   | 517   | 404   | 309   | 162   | 109   | 110   | 200   | 282   | 417   | 527  |
|                                                                                  | HDD18.3     | 7149   | 831   | 763   | 776   | 654   | 568   | 407   | 355   | 353   | 449   | 541   | 667   | 786  |
|                                                                                  | CDD10.0     | 35     | 0     | 0     | 0     | 0     | 0     | 6     | 13    | 15    | 1     | 0     | 0     | 0    |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.9    | 5.9   | 5.7   | 5.4   | 5.2   | 4.9   | 4.2   | 4.0   | 3.7   | 3.8   | 4.5   | 5.5   | 6.0  |
| Precipitation                                                                    | PrecAvg     | 1172   | 71    | 74    | 70    | 71    | 97    | 138   | 144   | 152   | 102   | 85    | 91    | 77   |
|                                                                                  | PrecMax     | 1583   | 178   | 250   | 160   | 169   | 164   | 263   | 236   | 265   | 189   | 199   | 258   | 214  |
|                                                                                  | PrecMin     | 923    | 7     | 18    | 13    | 15    | 32    | 66    | 76    | 37    | 38    | 5     | 0     | 6    |
|                                                                                  | PrecStd     | 148    | 46    | 51    | 37    | 31    | 40    | 47    | 39    | 47    | 40    | 41    | 52    | 49   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 2.7   | 3.7   | 3.7   | 6.1   | 10.7  | 15.9  | 16.9  | 17.2  | 13.6  | 11.4  | 7.5   | 3.8  |
|                                                                                  |             | MCWB   | -3.0  | -3.6  | -3.0  | 1.0   | 5.2   | 8.0   | 9.4   | 9.4   | 6.6   | 4.1   | 0.7   | -3.2 |
|                                                                                  | 2%          | DB     | 1.2   | 1.8   | 1.9   | 4.6   | 8.5   | 13.7  | 15.0  | 15.4  | 11.6  | 9.5   | 5.2   | 2.1  |
|                                                                                  |             | MCWB   | -3.8  | -3.9  | -3.4  | 0.2   | 4.1   | 7.8   | 9.0   | 8.9   | 6.1   | 3.4   | -0.6  | -4.0 |
|                                                                                  | 5%          | DB     | -0.7  | -0.5  | 0.5   | 3.1   | 6.9   | 12.1  | 13.4  | 13.6  | 10.0  | 8.0   | 3.7   | 0.7  |
|                                                                                  |             | MCWB   | -4.5  | -4.8  | -3.8  | -0.1  | 3.3   | 7.3   | 8.6   | 8.4   | 5.5   | 2.5   | -1.3  | -4.4 |
|                                                                                  | 10%         | DB     | -2.3  | -2.4  | -0.9  | 1.6   | 5.4   | 10.5  | 11.9  | 12.0  | 8.6   | 6.6   | 2.3   | -0.8 |
|                                                                                  |             | MCWB   | -5.2  | -5.5  | -4.1  | -1.0  | 2.3   | 6.7   | 7.8   | 7.7   | 4.8   | 1.8   | -2.0  | -4.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -1.0  | -1.3  | -0.8  | 2.1   | 6.2   | 10.0  | 11.2  | 11.2  | 8.4   | 5.9   | 2.7   | -0.9 |
|                                                                                  |             | MCDB   | 1.1   | 1.0   | 1.1   | 4.3   | 8.8   | 13.0  | 14.6  | 14.4  | 11.5  | 9.3   | 5.7   | 0.0  |
|                                                                                  | 2%          | WB     | -2.4  | -2.9  | -1.9  | 1.2   | 4.8   | 8.7   | 10.0  | 10.0  | 7.2   | 4.6   | 1.0   | -1.9 |
|                                                                                  |             | MCDB   | -0.8  | -1.0  | -0.2  | 3.3   | 7.5   | 12.1  | 13.6  | 13.5  | 10.0  | 7.6   | 3.5   | -0.4 |
|                                                                                  | 5%          | WB     | -3.5  | -4.0  | -2.7  | 0.3   | 3.7   | 7.9   | 9.0   | 9.0   | 6.3   | 3.7   | -0.1  | -2.8 |
|                                                                                  |             | MCDB   | -1.6  | -1.3  | -1.1  | 2.6   | 6.2   | 11.1  | 12.4  | 12.6  | 8.9   | 6.6   | 1.8   | -1.1 |
|                                                                                  | 10%         | WB     | -4.7  | -5.1  | -3.5  | -0.8  | 2.7   | 7.0   | 8.2   | 8.1   | 5.4   | 2.8   | -1.0  | -3.8 |
|                                                                                  |             | MCDB   | -2.6  | -2.3  | -1.6  | 1.3   | 4.9   | 10.0  | 11.2  | 11.3  | 7.8   | 5.4   | 0.9   | -1.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.7   | 4.6   | 4.6   | 4.5   | 4.5   | 5.2   | 6.0   | 6.0   | 5.3   | 5.0   | 4.4   | 4.7  |
|                                                                                  |             | MCDBR  | 5.4   | 5.3   | 4.9   | 4.9   | 5.7   | 6.8   | 7.6   | 7.7   | 7.0   | 6.2   | 5.0   | 5.2  |
|                                                                                  |             | MCWBR  | 3.7   | 3.3   | 3.3   | 3.5   | 4.0   | 4.4   | 4.7   | 4.7   | 5.0   | 4.4   | 3.5   | 3.4  |
|                                                                                  | 5% WB       | MCDBR  | 5.3   | 5.4   | 4.3   | 4.5   | 5.3   | 6.4   | 7.1   | 7.3   | 6.3   | 5.6   | 4.5   | 4.9  |
|                                                                                  |             | MCWBR  | 4.8   | 4.4   | 3.5   | 3.4   | 3.8   | 4.3   | 4.7   | 4.7   | 4.5   | 4.2   | 3.8   | 4.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.195  | 0.209 | 0.236 | 0.267 | 0.280 | 0.287 | 0.287 | 0.283 | 0.263 | 0.244 | 0.218 | 0.194 |      |
|                                                                                  | taud        | 2.439  | 2.405 | 2.399 | 2.455 | 2.526 | 2.541 | 2.537 | 2.568 | 2.622 | 2.651 | 2.594 | 2.475 |      |
|                                                                                  | Ebn at Noon | 959    | 1001  | 1008  | 996   | 986   | 976   | 972   | 967   | 968   | 950   | 924   | 924   |      |
|                                                                                  | Edh at Noon | 63     | 82    | 98    | 103   | 100   | 99    | 98    | 91    | 78    | 65    | 55    | 54    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.48   | 2.42  | 3.62  | 4.43  | 4.85  | 5.62  | 5.58  | 4.86  | 3.74  | 2.47  | 1.48  | 1.23  |      |
|                                                                                  | RadStd      | 0.17   | 0.29  | 0.48  | 0.67  | 0.50  | 0.50  | 0.48  | 0.46  | 0.38  | 0.29  | 0.22  | 0.16  |      |

## Historical Trends

|                          | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|--------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                          |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only             | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
|                          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (113 neighbors) | N/A   | N/A     | N/A    | +0.66   | +0.41 | +0.34 | N/A         | N/A     | +58     | +15     |

Nomenclature: See separate page